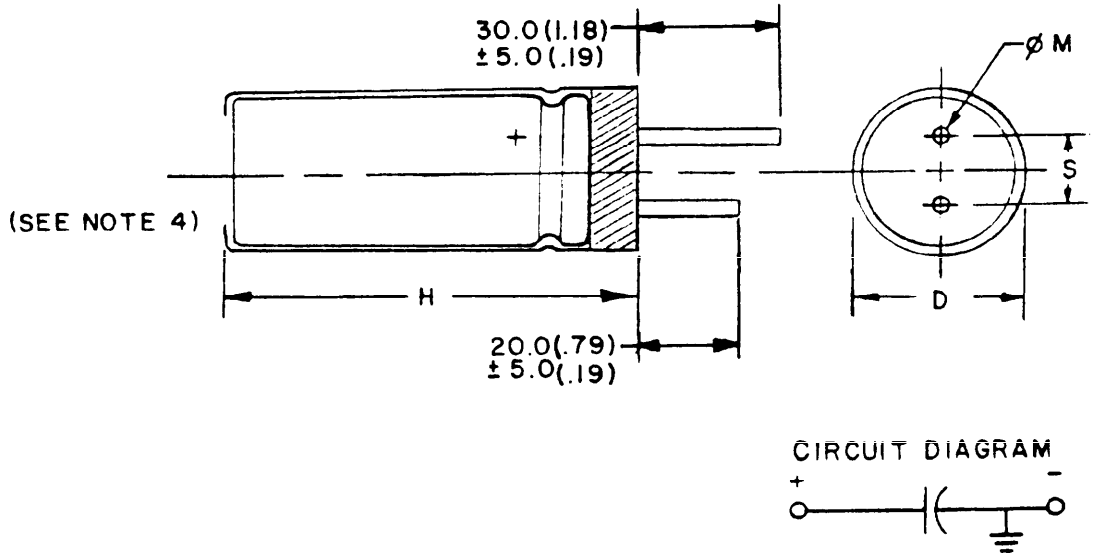
MILITARY SPECIFICATION SHEET

CAPACITORS, FIXED, ELECTROLYTIC, DOUBLE LAYER, CARBON, (POLARIZED),

STYLE DLO3 (METRIC)

This specification sheet is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the capacitors described herein shall consist of this specification sheet and the latest issue of DOD-C-29501.




CIRCUIT DIAGRAM




NOTES:
1. Dimensions are in millimeters.
2. Inch equivalents are given for general information only.
3. Inch equivalents are in parentheses.
4. Insulating sleeve shall overlap the end of the capacitor body.

FIGURE 1. Style DLO3 dimensions and configuration.


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DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

REQUIREMENTS:

Design and construction:

Dimensions and configuration: See figure 1.

Case material: Metal.

Leads: Solder-coated metal.

DC rated voltage: 1.8 volts.

Operating temperature: -40°C to +70°C.

DC leakage (DCL): See table I.

TABLE I. Style DL03 dimensions and characteristics.

| Military part number 1/ | Capacitance | Capacitance tolerance | Rated dc voltage | DC leakage (max) | $R_C$ | ESR | D ±0.5 | H ±2.0 | S ±0.5 | M ±0.05 |
|---|---|---|---|---|---|---|---|---|---|---|
| D29501A03 | Farads | Percent | Volts | µA | Ohms | Ohms | mm | mm | mm | mm |
| A334V | 0.33 | -20,+80 | 1.8 | 120 | 100 | 2.7 | 7.0 | 24.0 | 2.5 | 0.7 |
| A105V | 1.0 | " | " | 180 | 10 | 1.7 | 8.5 | 24.0 | 3.5 | 0.7 |
| A335V | 3.3 | " | " | 280 | 10 | 1.7 | 13.0 | 27.0 | 5.0 | 0.8 |
| A475V | 4.7 | " | " | 440 | 10 | 1.7 | 13.0 | 26.0 | 5.0 | 0.8 |
| A106V | 10.0 | " | " | 820 | 1 | 0.8 | 13.0 | 37.0 | 5.0 | 0.8 |

1/ The complete part number will be D29501A03A334V.

Capacitance: See table I.

Capacitance tolerance: -20, +80 percent.

ESR: See table I.

Thermal shock:

DCL: In accordance with DOD-C-29501.
ESR: In accordance with DOD-C-29501.

Resistance to soldering heat:

CAP: In accordance with DOD-C-29501.
ESR: In accordance with DOD-C-29501.

Moisture resistance:

CAP: In accordance with DOD-C-29501.
ESR: In accordance with DOD-C-29501.

Stability at low and high temperatures:

Step 1 (+25°C):

DCL: See table I.
CAP: Within tolerance.
ESR: See table I.

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Step 2 (-40°C):

DCL: Shall change not more than -200 percent from the initial step 1 measured value.
CAP: Shall change not more than -50 percent from the step 1 measured value.
ESR: Shall change not more than 400 percent from the table I measured value.

Step 3 (+25°C):

DCL: See table I.
CAP: Shall change not more than ±20 percent from the step 1 measured value.
ESR: See table I.

Step 4 (+70°C):

DCL: Shall change not more than 300 percent from the table I measured value.
CAP: Shall change not more than +150 percent from the step 1 measured value.
ESR: See table I.

Step 5 (+25°C):

DCL: See table I.
CAP: Shall change not more than ±20 percent from the step 1 measured value.
ESR: See table I.

Surge voltage: In accordance with DOD-C-29501.

DCL: Shall change not more than ±20 percent.
CAP: Within 20 percent of the initial measured value.
ESR: See table I.

Life: In accordance with DOD-C-29501.

DCL: Shall change not more than 200 percent of the table I value.
CAP: Shall change not more than -30 percent of the initial measured value.
ESR: Shall change not more than 200 percent of the table I value.

Resistance to solvents:

DCL: In accordance with DOD-C-29501.
ESR: In accordance with DOD-C-29501.

Custodians:
Army - ER
Navy - EC
Air Force - 11

Review activities:
Army - AV
Navy - SH
Air Force - 17, 85, 99
DLA - ES

User activities:
Army - MI
Navy - AS, CG, MC, OS
Air Force - 19

Preparing activity:
Navy - EC

Agent:
DLA - ES

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